Monday Coffee

City Expansion Analysis

Tool Used: PostgreSQL | Power BI

Prepared by: Prasiddhi Gupta

Data Analyst

1. Project Overview

Monday Coffee started selling **online in January 2023** and the response from customers has been really good across multiple Indian cities. Now the company wants to take the **next step and open physical coffee shops**.

My job was to look at the sales data, understand which cities are performing well, and give a clear recommendation on where the company should open its first 3 stores.

Goal	Find top 3 cities for physical store expansion
Cities Looked At	14
Total Sales Records	10,388
Tool Used	PostgreSQL — pgAdmin 4, Power BI

2. What I Was Trying to Find Out

Before writing a single query I sat down and thought about what actually matters when deciding where to open a store. I came up with these questions:

-- Where are customers already buying the most online?
-- Where are customers the most happy with the product?
-- After paying rent will the store actually make money?
-- Is there still room to grow in that city or is the market already saturated?
-- Are there cities that look good on paper but should actually be avoided?
These questions guided my entire analysis.

3. About the Data

city — Information about 14 cities including population, estimated rent and city rank
customers — 497 customers and which city they belong to
products — 28 coffee products with their prices
sales — 10,388 sales records with date, product, customer, amount and rating

The tables were connected through foreign keys which is why I had to create the schema first before importing the data.

4. Analysis — Queries and What I Found

Query 1 — Which city makes the most revenue?

This was my starting point. I wanted to know where customers are already spending the most money.

```sql
SELECT
ci.city_name,
ROUND(SUM(s.total)::numeric, 2) AS total_revenue
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name
ORDER BY total_revenue DESC;
```

Pune came first at ₹12,58,290 which was quite a big lead over the others. Chennai was second at ₹9,44,120 and Bangalore third at ₹8,60,110.

Query 2 — How many orders is each city placing?

Revenue can sometimes be misleading if one city has very few high value orders. So I checked order count separately.

```Sql
SELECT
  ci.city_name,
COUNT(s.sale_id) AS total_orders
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name
ORDER BY total_orders DESC;
```

The same top 3 cities held their position here as well which gave me more confidence in the data.

Query 4 — Where are customers most satisfied?

Ratings are really important for expansion. If customers online are already giving low ratings then opening a physical store there is risky.

```sql
Sql
SELECT
 ci.city_name,
 ROUND(AVG(s.rating)::numeric, 2) AS avg_rating
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name
ORDER BY avg_rating DESC;
```

Chennai had the highest rating of 4.52 across all 14 cities. Bangalore was 4.48 and Pune 4.47. This was a strong signal for all three cities.

Query 6 — Which cities have repeat customers?

Repeat customers show that people actually like the product enough to come back. This is a green flag for opening a physical store.

```sql
Sql
SELECT
   ci.city_name,
      COUNT(DISTINCT s.customer_id) AS repeat_customers
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name
HAVING COUNT(s.sale_id) > 1
ORDER BY repeat_customers DESC;
```

Query 8 — Which city will be most profitable after rent?

This was a key query. High revenue means nothing if rent eats it all up.

```sql
Sql
SELECT
```

ci.city_name,
ROUND(SUM(s.total)::numeric, 2) AS revenue,
ci.estimated_rent,
ROUND(
(SUM(s.total) - ci.estimated_rent)::numeric, 2
) AS estimated_profit
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name, ci.estimated_rent
ORDER BY estimated_profit DESC;

Pune profit ₹12,42,990 — Chennai ₹9,27,020 — Bangalore ₹8,30,410

Query 10 — How are monthly sales trending?

I wanted to see if sales are growing over time or declining in each city.

Sql
SELECT
ci.city_name,
TO_CHAR(s.sale_date, 'YYYY-MM') AS month,
ROUND(SUM(s.total)::numeric, 2) AS monthly_sales
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name, month
ORDER BY ci.city_name, month;

Query 11 — Population vs Revenue

Large population cities should ideally have large revenue. Where this is not happening it means the market is not ready yet.

Sql
SELECT
ci.city_name,
ci.population,
ROUND(SUM(s.total)::numeric, 2) AS revenue
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id

JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name, ci.population
ORDER BY revenue DESC;

Mumbai has 2 crore population but only ₹2,35,000 revenue. That was a red flag.

Query 12 — What products sell best in each city?

Knowing the popular products helps in planning the store menu.

```
Sql
SELECT
 ci.city_name,
 p.product_name,
 COUNT(*) AS total_orders
FROM sales s
JOIN products p ON s.product_id = p.product_id
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name, p.product_name
ORDER BY ci.city_name, total_orders DESC;
```

Query 7 — Revenue per customer by city

This tells me how valuable each customer is in a city.

```
Sql
SELECT
 ci.city_name,
 ROUND(
  SUM(s.total)::numeric /
COUNT(DISTINCT c.customer_id), 2
 ) AS revenue_per_customer
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name
ORDER BY revenue_per_customer DESC;
```

Query 15 — Final Recommendation Query

This was the final query that brought everything together.

```
Sql
SELECT
 ci.city_name,
 ROUND(SUM(s.total)::numeric,2) AS revenue,
 COUNT(DISTINCT c.customer_id) AS customers, ROUND(AVG(s.rating)::numeric,2) AS
avg_rating,
  ci.population,
 ci.estimated_rent,
ROUND((SUM(s.total) / ci.population)::numeric, 4)
 AS revenue_per_person,
 ROUND((SUM(s.total) - ci.estimated_rent)::numeric, 2)
AS estimated_profit
FROM sales s
JOIN customers c ON s.customer_id = c.customer_id
JOIN city ci ON c.city_id = ci.city_id
GROUP BY ci.city_name, ci.population, ci.estimated_rent
ORDER BY revenue DESC, avg_rating DESC, customers DESC
LIMIT 3;
```

Result:

Pune — Revenue ₹12,58,290 — Customers 52 — Rating 4.47 — Rent ₹15,300 — Profit ₹12,42,990

Chennai — Revenue ₹9,44,120 — Customers 42 — Rating 4.52 — Rent ₹17,100 — Profit ₹9,27,020

Bangalore — Revenue ₹8,60,110 — Customers 39 — Rating 4.48 — Rent ₹29,700 — Profit ₹8,30,410

5. What the PostgreSQL Data Told Me

Revenue picture is clear Pune, Chennai and Bangalore are in a completely different league compared to the other 11 cities. The gap between Bangalore at third and Jaipur at fourth is significant enough to make the top 3 an easy choice from a revenue standpoint.

Ratings sealed the decision Jaipur and Delhi which were 4th and 5th in revenue had ratings of only 3.51. That is a big drop from the top 3 which are all above 4.47. Low online ratings mean customers are not fully happy — opening a physical store in such a city is a gamble.

Mumbai was a surprise Mumbai has 2 crore population — the most of any city in the dataset. But its revenue is only ₹2,35,000 which is lower than others cities with far smaller populations like Kanpur and Surat. The market in Mumbai is clearly not ready for a physical store yet and its rent of ₹31,500 is the highest of all 14 cities.

Rent matters more than I expected Bangalore has the highest rent among the top 3 at ₹29,700. Initially I was worried this would hurt its case. But when I looked at the estimated profit it was still ₹8,30,410 which is very healthy. The revenue is strong enough to absorb the higher rent.

6. Power BI Dashboard Tool Used: Power BI Desktop

After completing the SQL analysis I connected Power BI directly to the PostgreSQL database and built an interactive single page dashboard to visualize all the findings in one place.

How I Connected PostgreSQL to Power BI

Get Data → PostgreSQL Database → Server: localhost → Database: monday_coffee_db → Load all 4 tables directly into Power BI. No CSV export needed — live connection se data directly aaya.

Dashboard Design

I used a custom Canva background with a coffee theme to match the Monday Coffee brand. **Canvas size was set to 3000 x 1600** for a wide format layout. **Two slicers were added — City Slicer and Product Slicer** — so anyone viewing the dashboard can filter all charts by a specific city or product instantly.

KPI Cards

Five KPI cards at the top give an instant overview:

Total Revenue — 60,70,190

Total Orders — 10,388

Total Customers — 497

Average Rating — 4.0

Total Cities — 14

Charts and What They Show

Revenue by City — Horizontal bar chart sorted descending. **Pune leads at 1.26M followed by Chennai 0.94M and Bangalore 0.86M**. The gap between top 3 and the rest is clearly visible.

Rating by City — Bar chart showing customer satisfaction. **Chennai tops at 4.52, Bangalore 4.48, Pune 4.47. Cities like Hyderabad and Kanpur** fall below 3.5 which is a red flag for expansion.

Estimated Profit by City — Shows profit after deducting rent. **Pune leads at 1.24M estimated profit.** This chart directly answers whether **a city is worth opening a store in.**

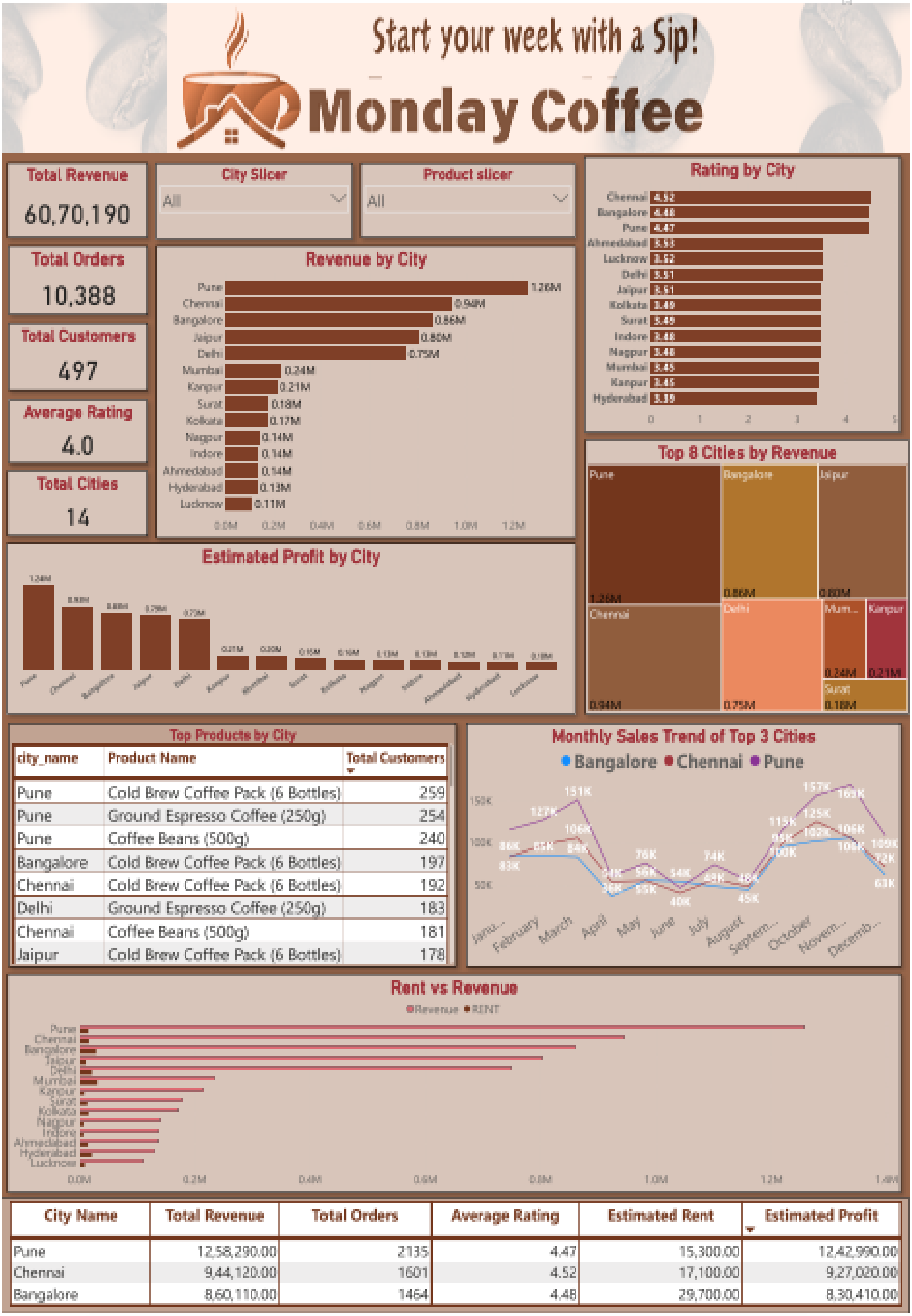
Rent vs Revenue — Clustered bar chart comparing rent burden vs actual revenue for all 14 cities. Mumbai stands out here — high rent but very low revenue — worst combination.

Monthly Sales Trend of Top 3 Cities — Line chart showing Bangalore, Chennai and Pune month by month. **Pune consistently leads and shows an upward trend** in the latter months which is a positive signal.

Top Products by City — Table showing best selling products in each city. **Cold Brew Coffee Pack and Ground Espresso Coffee are top sellers** across most cities — **useful for planning store menu.**

Top 8 Cities by Revenue — Treemap giving a quick visual comparison of top 8 cities by revenue size.

Final Recommendation Table — Clean table showing only **Pune, Chennai and Bangalore** with Revenue, Total Orders, Average Rating, Estimated Rent and Estimated Profit side by side.



Top Products by City

city_name	Product Name	Total Customers
Pune	Cold Brew Coffee Pack (6 Bottles)	259
Pune	Ground Espresso Coffee (250g)	254
Pune	Coffee Beans (500g)	240
Bangalore	Cold Brew Coffee Pack (6 Bottles)	197
Chennai	Cold Brew Coffee Pack (6 Bottles)	192
Delhi	Ground Espresso Coffee (250g)	183
Chennai	Coffee Beans (500g)	181
Jaipur	Cold Brew Coffee Pack (6 Bottles)	178

City Name	Total Revenue	Total Orders	Average Rating	Estimated Rent	Estimated Profit
Pune	12,58,290.00	2135	4.47	15,300.00	12,42,990.00
Chennai	9,44,120.00	1601	4.52	17,100.00	9,27,020.00
Bangalore	8,60,110.00	1464	4.48	29,700.00	8,30,410.00

7. My Recommendation

City 1 — Pune

Pune is the strongest choice by every measure. Highest revenue, most customers, lowest rent among the top 3 and the best profit. Customers here are already buying online regularly which means there is real demand. A physical store will make it even more convenient for them.

What to do next: Look for store locations in areas with high young population and office density in Pune.

City 2 — Chennai

Chennai customers gave the highest average rating of 4.52 across all 14 cities. This tells me they genuinely love the product. Happy customers talk about their experience and bring in new customers — especially important when opening a new physical store. Revenue and profit numbers are solid as well.

What to do next: Focus on areas with high young professional footfall in Chennai.

City 3 — Bangalore

Bangalore is India's tech and startup hub. Young professionals with good salaries and a strong coffee culture make it a natural fit for Monday Coffee. The revenue and profit numbers back this up. Yes the rent is higher but the market more than justifies it.

What to do next: Target tech parks, co-working spaces and startup neighborhoods in Bangalore.

Cities I would not recommend right now:

Jaipur and Delhi look attractive because they have many customers — 69 and 68 respectively. But their ratings of 3.51 tell a different story. The product experience online is not great in these cities and taking that experience into a physical store could hurt the brand.

Mumbai feels like it should be on the list given its size. But the data does not support it. The revenue is too low for such a large city and the rent is the highest of all 14. It is a city to watch for the future — not for the first wave of expansion.

7. Conclusion

This was a really interesting project because the answer was not always obvious from looking at one number. I had to look at revenue, ratings, rent, population and profit together before making a call.

The biggest learning for me was that a city with a huge population does not automatically mean a good market. Mumbai is the perfect example of that. And a city with many customers does not mean they are happy customers — Jaipur showed me that.

In the end Pune, Chennai and Bangalore stood out clearly — not because of one strong metric but because they were consistently good across all the metrics that matter for a physical store expansion.